

#14Days14Chapters

DAY 12

[Aim: 100/100 in Maths]

QUADRATIC EQUATIONS

Quadratic Polynomial

↳ $D=2$

~~eg~~

$$3x^2 + 4x - 2$$

Quadratic Equation $(\Leftrightarrow)$

~~eg~~ $\boxed{4x^2 - 2x + 3 = 0}$

$\boxed{D=2}$

#LP : Identify which of the following are Quadratic Equation ?

(i) $x^2 - 6x + 3 = 0$

→ $D = 2$

✓✓

(ii) $x^2 + 1/x^2 = 3$

~~$x^2 + \frac{1}{x^2} = 3$~~

~~$x^2 + x^{-2} = 3$~~

~~$D = 2$~~

~~poly~~

✓✓

→ $x^2 + \frac{1}{x^2} = 3$

⇒ $\frac{x^4 + 1}{x^2} = 3$

⇒ $x^4 + 1 = 3x^2$

→ $x^4 - 3x^2 + 1 = 0$

$D = 4$ Quad

$$(iii) (2x+1)(3x+2) = 6(x-1)(x-2)$$

~~D=1~~
~~Quad.~~

$$\rightarrow 6x^2 + 4x + 3x + 2 = (6x - 6)(x - 2)$$

$$\rightarrow \cancel{6x^2} + 7x + 2 = \cancel{6x^2} - 12x - 6x + 12$$

$$\rightarrow 7x + 2 = -18x + 12$$

$$\rightarrow 7x + 18x = 12 - 2$$

$$\rightarrow \cancel{25}x = \cancel{10}2$$

$$x = \frac{2}{5}$$

D=1 quad.

$$(iv) (x+2)^3 = x^3 - 4$$

$$\Rightarrow (x+2)^3 + 3(x)(2)(x+2) = x^3 - 4$$

$$\rightarrow \cancel{x^3} + 8 + 6x(x+2) = \cancel{x^3} - 4$$

$$\rightarrow 8 + 6x^2 + 12x = -4$$

$$\rightarrow \boxed{6x^2 + 12x + 12 = 0}$$

D=2
quad ✓

Zeroes of Quadratic Equation:

→ /roots /solution.

→ x की वो value जिससे $quad. = 0$ हो जाए।

$$ax^2 + bx + c = 0$$

✓ Splitting the middle term
Method

✓ Discriminant Method
(Quadratic formula)

$$x^2 + 5x + 6 = 0$$

$$x^2 + 2x + 3x + 6 = 0$$

$$x(x+2) + 3(x+2) = 0$$

$$(x+2)(x+3) = 0$$

$$x+2=0$$

$$x = -2$$

$$x+3=0$$

$$x = -3$$

$\therefore -2$ & -3 are zeros of this quad. eqⁿ.

$$p+q=5$$

$$pq=6$$

2	6
3	3
	1

Splitting the middle term (Factorisation):

#LP: $8x^2 - 22x - 21 = 0$ ← 3 terms

$8x^2 - 28x + 6x - 21 = 0$ ← 4 terms

$4x(2x - 7) + 3(2x - 7) = 0$ ← common

$(2x - 7)[4x + 3] = 0$

$2x - 7 = 0$

$x = 7/2$

$4x + 3 = 0$

$x = -3/4$

$p + q = -22$

$pq = -168$

Factor tree for 168:

2	168
2	84
2	42
3	21
7	7

Factors: 6, 28, 7, 7

#LP: Solve for x:

$$x^2 - (\sqrt{3} + 1)x + \sqrt{3} = 0$$

$$\underbrace{x^2 - \sqrt{3}x} - \underbrace{x + \sqrt{3}} = 0 \quad \leftarrow \text{4 terms}$$

$$\underline{x}(\underbrace{x - \sqrt{3}}) - \underline{1}(\underbrace{x - \sqrt{3}}) = 0$$

$$(x - \sqrt{3})[x - 1] = 0$$

$$\begin{array}{l|l} x - \sqrt{3} = 0 & x - 1 = 0 \\ \hline x = \sqrt{3} & x = 1 \end{array}$$

#LP: Solve the quadratic equation: $x^2 - 2ax + (a^2 - b^2) = 0$ for x . (2022)

$$x^2 - 2ax + (a^2 - b^2) = 0$$

$$\Rightarrow x^2 - 2ax + (a+b)(a-b) = 0$$

$$\Rightarrow x^2 - (a+b)x - (a-b)x + (a+b)(a-b) = 0$$

$$\Rightarrow x[x - (a+b)] - (a-b)[x - (a+b)] = 0$$

$$[x - (a+b)] [x - (a-b)] = 0$$

$$x - (a+b) = 0$$

$$\boxed{x = a+b} \checkmark$$

$$x - (a-b) = 0$$

$$\boxed{x = a-b} \checkmark$$

$$p+q = -2a$$

$$pq = (a+b)(a-b)$$

$$-(a+b) - (a-b)$$

$$\Rightarrow -a - b - a + b$$

$$\Rightarrow -2a$$

#LP: Solve the quadratic equation for x:

$$x^2 - 2ax - (4b^2 - a^2) = 0 \quad (\text{Term II, 2021-22, AI 2015})$$

$$x^2 - 2ax - (4b^2 - a^2) = 0$$

$$x^2 - 2ax - [(2b)^2 - (a)^2] = 0$$

$$x^2 - 2ax - (2b+a)(2b-a) = 0$$

$$x^2 - (2b+a)x + (2b-a)x - (2b+a)(2b-a) = 0$$

$$x[x - (2b+a)] + (2b-a)[x - (2b+a)] = 0$$

$$[x - (2b+a)](x + (2b-a)) = 0$$

$$p+q = -2a$$

$$pq = (2b+a)(2b-a)$$

$$p = (2b+a)$$

$$q = -(2b-a)$$

$$-(2b+a) + (2b-a)$$

$$\Rightarrow -2b - a + 2b - a$$

$$\Rightarrow -2a$$

$$x - (2b + a) = 0$$

$$\boxed{x = 2b + a}$$

$$x + (2b - a) = 0$$

$$\boxed{x = -(2b - a)}$$

#LP:

Solve for x : $\frac{1}{x-3} - \frac{1}{x+5} = \frac{1}{6}, x \neq 3, -5$

$$\Rightarrow \frac{1}{x-3} - \frac{1}{x+5} = \frac{1}{6}$$

$$\Rightarrow \frac{1(x+5) - 1(x-3)}{(x-3)(x+5)} = \frac{1}{6}$$

$$\Rightarrow \frac{\cancel{x}+5-\cancel{x}+3}{x^2+5x-3x-15} = \frac{1}{6}$$

$$\Rightarrow \frac{8}{x^2+2x-15} = \frac{1}{6}$$

$$\Rightarrow 48 = x^2 + 2x - 15$$

$$0 = x^2 + 2x - 15 - 48$$

$$x^2 + 2x - 63 = 0$$

$$x^2 - 7x + 9x - 63 = 0$$

$$x(x-7) + 9(x-7) = 0$$

$$(x-7)(x+9) = 0$$

$$\begin{array}{l|l} x-7=0 & x+9=0 \\ \hline \boxed{x=7} & \boxed{x=-9} \end{array}$$

$$p+q=2$$

$$pq=-63$$

$$\begin{array}{r|l} 3 & 63 \\ \hline 9 & 21 \\ \hline 3 & 7 \\ \hline 1 & 1 \end{array}$$

#LP: Solve the following quadratic equation:

$$\frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x}, a+b \neq 0$$

$$\Rightarrow \frac{1}{a+b+x} - \frac{1}{x} = \frac{1}{a} + \frac{1}{b}$$

$$\Rightarrow \frac{1(x) - 1(a+b+x)}{(a+b+x)(x)} = \frac{1(b) + 1(a)}{ab}$$

$$\Rightarrow \frac{\cancel{x} - a - b - \cancel{x}}{(a+b+x)(x)} = \frac{a+b}{ab}$$

$$\Rightarrow \frac{-\cancel{(a+b)}}{ax+bx+x^2} = \frac{\cancel{a+b}}{ab}$$

$$\Rightarrow \frac{-1}{ax+bx+x^2} = \frac{1}{ab}$$

$$\Rightarrow -ab = ax+bx+x^2$$

$$\Rightarrow 0 = x^2+ax+bx+ab$$

$$\boxed{x^2+ax+bx+ab=0} \leftarrow \text{4 terms}$$

$$\underline{x(x+a)} + \underline{b(x+a)} = 0$$

$$(x+a)(x+b) = 0$$

$$x+a=0 \quad | \quad x+b=0$$

$$\boxed{x=-a} \quad | \quad \boxed{x=-b}$$

Discriminant method:

$$ax^2 + bx + c = 0$$

Quadratic
Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\left(x = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \right)$$

$$\left(x = \frac{-b - \sqrt{b^2 - 4ac}}{2a} \right)$$

(ਸ਼੍ਰੀ ਧਰਮਾਰਜ ਜੀ)

$$x^2 + 5x + 6 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(5) \pm \sqrt{(5)^2 - 4(1)(6)}}{2(1)}$$

$$x = \frac{-5 \pm \sqrt{25 - 24}}{2}$$

$$x = \frac{-5 \pm \sqrt{1}}{2}$$

$$x = \frac{-5 \pm 1}{2} \Rightarrow \frac{-5+1}{2} = \frac{-4}{2} = -2$$

$$x = \frac{-5 \pm 1}{2} \Rightarrow \frac{-5-1}{2} = \frac{-6}{2} = -3$$

close 2026.

Nature of Roots:-

Key Property

$$D = b^2 - 4ac$$

- $D > 0 \rightarrow$ roots are real and distinct
अलग-अलग
- $D = 0 \rightarrow$ roots are real & equal
- $D < 0 \rightarrow$ unreal roots

Drona JEE
Drona NEET

Drona JEE/NEET → 11th + School

(P, C, M, B, E)
Prarambh
Sci
(free)

Sci
Comm
Hum.

Prarambh

Early 500

17th ⇒ Drona JEE/
Drona NEET } → Prarambhi Sci (free)

=

Early 500

#LP: Find the nature of roots without solving.

$$2x^2 - 5x + 3 = 0$$

$$D = b^2 - 4ac$$

$$= (-5)^2 - 4(2)(3)$$

$$= 25 - 24$$

$$D = 1 \rightarrow > 0$$

$\therefore$ Real & distinct roots.

#LP: The value(s) of k for which the quadratic equation $2x^2 + kx + 2 = 0$ has equal roots, is

(a) 4

☒ (b) ± 4

(c) -4

(d) 0 (2020, 1 Mark)

Equal roots $\rightarrow D = 0$

$$b^2 - 4ac = 0$$

$$(k)^2 - 4(2)(2) = 0$$

$$k^2 - 16 = 0$$

$$k^2 - (4)^2 = 0$$

$$(k+4)(k-4) = 0$$

$$k+4=0$$

$$\underline{k = -4}$$

$$k-4=0$$

$$\underline{k = 4}$$

$$2x^2 - 4x + 2 = 0$$

$$2x^2 + 4x + 2 = 0$$

#LP: : If the roots of quadratic equation $4x^2 - 5x + k = 0$ are real and equal, then the value of k is (CBSE

2024)

(a) $5/4$

☒ (b) $25/16$

(c) $-\frac{5}{4}$

(d) $-\frac{25}{16}$

$\frac{25}{16}$

$D = 0$

$b^2 - 4ac = 0$

$(-5)^2 - 4(4)(k) = 0$

$25 - 16k = 0$

$25 = 16k$

$\frac{25}{16} = k$

#LP: Find the value of m for which the quadratic equation $(m-1)x^2 + 2(m-1)x + 1 = 0$ has two real and equal roots. (2022)

Equal roots $\rightarrow D = 0$

$$b^2 - 4ac = 0$$

$$\Rightarrow [2(m-1)]^2 - 4(m-1)(1) = 0$$

$$\Rightarrow 4(m-1)^2 - 4(m-1) = 0$$

$$\Rightarrow 4(m-1)[(m-1)-1] = 0$$

$$\Rightarrow 4(m-1)(m-2) = 0$$

$$m-1=0$$

$$\boxed{m=1}$$

$$(1-1)x^2 + 2(1-1)x + 1 = 0$$

x^2 की term उजागरे

$$m-2=0$$

$$\boxed{m=2}$$

(A) 1 — 6%

~~(B) 2 — 14%~~

~~(C) 1, 2 — 65%~~

(D) NOTA — 14%

#LP: The quadratic equation $(1 + a^2)x^2 + 2abx + (b^2 - c^2) = 0$ has only one root. What is the value of $c^2(1 + a^2)$

~~Q. 17.~~ (2022)

$$Ax^2 + Bx + C = 0$$

both roots equal.

Equal roots $\rightarrow D = 0$ ✓

$$B^2 - 4AC = 0$$

$$\Rightarrow (2ab)^2 - 4(1+a^2)(b^2-c^2) = 0$$

$$\Rightarrow 4a^2b^2 - (4 + 4a^2)(b^2 - c^2) = 0$$

$$\Rightarrow 4a^2b^2 - [4b^2 - 4c^2 + 4a^2b^2 - 4a^2c^2] = 0$$

$$\Rightarrow \cancel{4a^2b^2} - 4b^2 + 4c^2 - \cancel{4a^2b^2} + 4a^2c^2 = 0$$

$$\Rightarrow -4b^2 + 4c^2 + 4a^2c^2 = 0$$

$$\Rightarrow -4(b^2 - c^2 - a^2c^2) = 0$$

$$b^2 - c^2 - a^2c^2 = 0$$

$$b^2 = c^2 + a^2c^2$$

$$\boxed{b^2 = c^2(1 + a^2)}$$

Ans $\rightarrow \underline{\underline{b^2}}$

~~K³B~~

$D=1 \longrightarrow \text{max } 1 \text{ root}$

$D=2 \longrightarrow \text{max } 2 \text{ roots}$

$D=3 \longrightarrow \underline{\underline{\text{max}}} \text{ } 3 \text{ roots}$

$D=4 \longrightarrow \text{max } 4 \text{ roots}$

;

;

;

$0^{\checkmark}, 1^{\checkmark}, 2^{\checkmark}, 3^{\times}$

#LP:

If the roots of the equation $(a^2 + b^2)x^2 - 2(ac + bd)x + (c^2 + d^2) = 0$ are equal, prove that $\frac{a}{b} = \frac{c}{d}$

$$Ax^2 + Bx + C = 0$$

[CBSE 2017]

$$D = 0$$

$$B^2 - 4AC = 0$$

$$[-2(ac + bd)]^2 - 4(a^2 + b^2)(c^2 + d^2) = 0$$

$$\Rightarrow (-2)^2 (ac + bd)^2 - [(4a^2 + 4b^2)(c^2 + d^2)] = 0$$

$$\Rightarrow 4[(a^2c^2 + b^2d^2 + 2ac)(bd)] - [4a^2c^2 + 4a^2d^2 + 4b^2c^2 + 4b^2d^2] = 0$$

$$\Rightarrow 4a^2c^2 + 4b^2d^2 + 8acbd - 4a^2c^2 - 4a^2d^2 - 4b^2c^2 - 4b^2d^2 = 0$$

$$\Rightarrow 8abcd - 4a^2d^2 - 4b^2c^2 = 0$$

$$\Rightarrow 4(2abcd - a^2d^2 - b^2c^2) = 0$$

$$\Rightarrow -a^2d^2 - b^2c^2 + 2abcd = 0$$

$$\Rightarrow -(a^2d^2 + b^2c^2 - 2abcd) = 0$$

$$\Rightarrow (ad)^2 + (bc)^2 - 2(ad)(bc) = 0$$

$$A^2 + B^2 - 2AB$$

$$\Rightarrow (A - B)^2$$

$$(ad - bc)^2 = 0$$

$$ad - bc = 0$$

$$ad = bc$$

$$\frac{a}{b} = \frac{c}{d}$$

#LP: Find the smallest value of p for which the quadratic equation has real roots. Hence, find the roots of the equation so obtained.

~~38~~

real & dist $\rightarrow D > 0$

real & eq. $\rightarrow D = 0$

Only real roots $\rightarrow D \geq 0$

$$x^2 - 2(p+1)x + p^2 = 0$$

Real roots $\rightarrow D \geq 0$

$$b^2 - 4ac \geq 0$$

$$[-2(p+1)]^2 - 4(1)(p^2) \geq 0$$

$$(-2)^2 (p+1)^2 - 4p^2 \geq 0$$

$$\rightarrow 4(p^2 + 1 + 2p) - 4p^2 \geq 0$$

$$\rightarrow 4\cancel{p^2} + 4 + 8p - 4\cancel{p^2} \geq 0$$

$$\rightarrow 8p + 4 \geq 0$$

$$4(2p+1) \geq 0$$

$$2p+1 \geq 0$$

$$2p \geq -1$$

$$p \geq -\frac{1}{2}$$

$$p \rightarrow -\frac{1}{2} \text{ or } 3\text{से बड़ा}$$

$\therefore$ Smallest value of $p = -\frac{1}{2}$

Word Problem:

$$d = s \times t$$

#LP: 49. A train travels at a certain average speed for a distance of 63 km and then travels a distance of 72 km at an average speed of 6 km/hr more than its original speed. If it takes 3 hours to complete total journey, what is the original average speed? (2018)

$$d = 63 \text{ km}$$

$$s = x \text{ km/hr.}$$

$$t_1 = \frac{d}{s}$$

$$t_1 = \frac{63}{x}$$

$$d' = 72 \text{ km}$$

$$s' = (x + 6) \text{ km/hr}$$

$$t_2 = \frac{d'}{s'}$$

$$t_2 = \frac{72}{x+6}$$

$$\text{Total time} = 3$$

$$t_1 + t_2 = 3$$

$$\frac{63}{x} + \frac{72}{x+6} = 3$$

$$\Rightarrow \left(\frac{7}{x} + \frac{8}{x+6} \right) = \frac{1}{3}$$

$$\Rightarrow \frac{7}{x} + \frac{8}{x+6} = \frac{1}{3}$$

$$\Rightarrow \frac{7(x+6) + 8x}{x(x+6)} = \frac{1}{3}$$

$$\Rightarrow \frac{7x + 42 + 8x}{x^2 + 6x} = \frac{1}{3}$$

$$\Rightarrow 3(15x + 42) = x^2 + 6x$$

$$\Rightarrow 45x + 126 = x^2 + 6x$$

$$\Rightarrow x^2 + 6x - 45x - 126 = 0$$

$$x^2 - 39x - 126 = 0$$

$$x^2 - 42x + 3x - 126 = 0$$

$$x(x-42) + 3(x-42) = 0$$

$$(x-42)(x+3) = 0$$

$$x-42=0 \quad | \quad x+3=0$$

$$x=42$$

$$x=-3$$

Speed cannot be -ve

$$\therefore \text{Speed} = 42 \text{ km/hr}$$

$$p+q = -39$$

$$pq = -126$$

2	126
3	63
3	21
7	7
	1

#LP: In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/hr from its usual speed and the time of the flight increased by 30 min. Find the scheduled duration of the flight. (2022-23)

let original speed = x km/hr

original time taken $\rightarrow y$ hr

$$d = s \times t$$

$$600 = x \times y$$

$$\frac{600}{y} = x$$

$$s' = (x - 200) \text{ km/hr}$$

$$t' = \left(y + \frac{1}{2}\right) \text{ hr}$$

d remains same

$$600 = s' \times t'$$

$$600 = (x - 200) \left(y + \frac{1}{2}\right)$$

$$600 = \left(\frac{600}{y} - 200\right) \left(y + \frac{1}{2}\right)$$

$$\cancel{300} = \cancel{200} \left(\frac{3}{y} - 1 \right) \left(y + \frac{1}{2} \right)$$

$$3 = \left(\cancel{\frac{3}{y}} + \frac{3}{2y} - y - \frac{1}{2} \right)$$

$$\cancel{3} = \cancel{3} + \frac{3}{2y} - y - \frac{1}{2}$$

$$\frac{1}{2} = \frac{3}{2y} - y$$

$$\cancel{1} = \frac{3 - 2y^2}{\cancel{2}y}$$

$$y = 3 - 2y^2$$

$$\boxed{2y^2 + y - 3 = 0}$$

$$2y^2 + 3y - 2y - 3 = 0$$

$$y(2y+3) - 1(2y+3) = 0$$

$$(y-1)(2y+3) = 0$$

$$y-1=0$$

$$\underline{y=1}$$

$$2y+3=0$$

$$y = -3/2$$

• original time = 1 hr, t cannot be -ve

$$p+q=1$$

$$p \cdot q = -6$$

$$\begin{array}{r|l} 2 & 6 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

#LP: Two pipes running together can fill a tank in $\frac{100}{9}$ minutes. If one pipe takes 5 minutes more than the other to fill the tank separately, find the time in which each pipe would fill the tank separately.

$$\text{Together time taken} = \frac{100}{9} \text{ min}$$

separately

Let time taken by big pipe to fill tank comple $\Rightarrow x$ min

x min $\rightarrow$ 1 work

1 min $\rightarrow \left(\frac{1}{x}\right)$ work

$\frac{100}{9}$ min $\rightarrow \left[\frac{100}{9} \left(\frac{1}{x}\right)\right]$ work

time taken by smaller pipe separately to fill tank

$\Rightarrow (x+5)$ min

$(x+5)$ min $\rightarrow$ 1 work

1 min $\rightarrow \frac{1}{x+5}$

$\frac{100}{9}$ min $\rightarrow \frac{100}{9} \left(\frac{1}{x+5}\right)$

tank fill $\Rightarrow$ 1 work

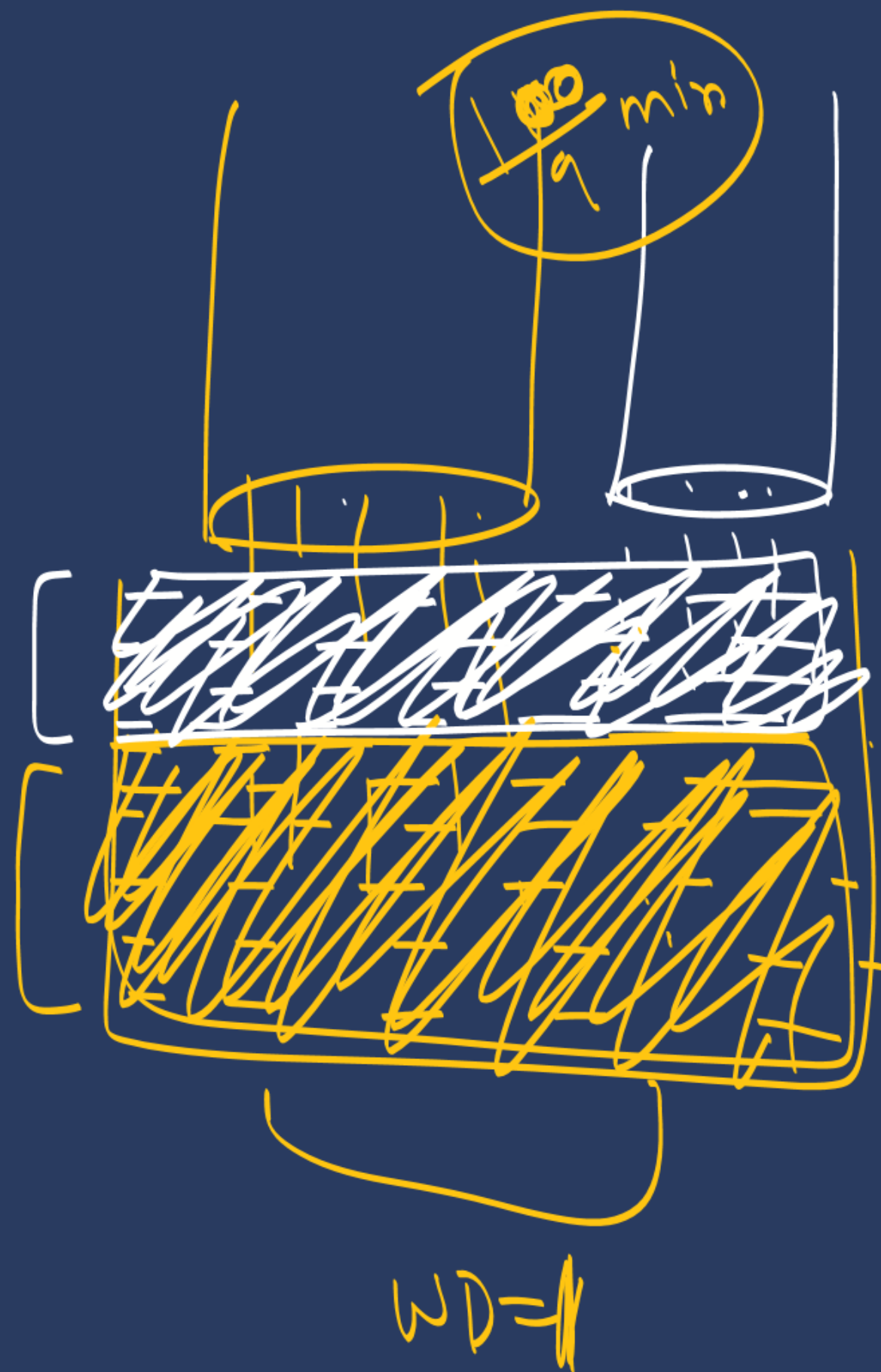


Together for $\frac{100}{9}$ min

$$\frac{(W \cdot D)_{\text{big}}}{\frac{100}{9} \left(\frac{1}{x} \right)} + \frac{(W \cdot D)_{\text{small}}}{\frac{100}{9} \left(\frac{1}{x+5} \right)} = 1$$

$$\frac{1}{9} \left(\frac{1}{x} + \frac{1}{x+5} \right) = 1$$

$$\frac{1}{x} + \frac{1}{x+5} = \frac{9}{1}$$



#1C3B

STILL WATER



Speed of stream (t) = 0

Speed of boat (s) = s km/hr.
in still water

UPSTREAM

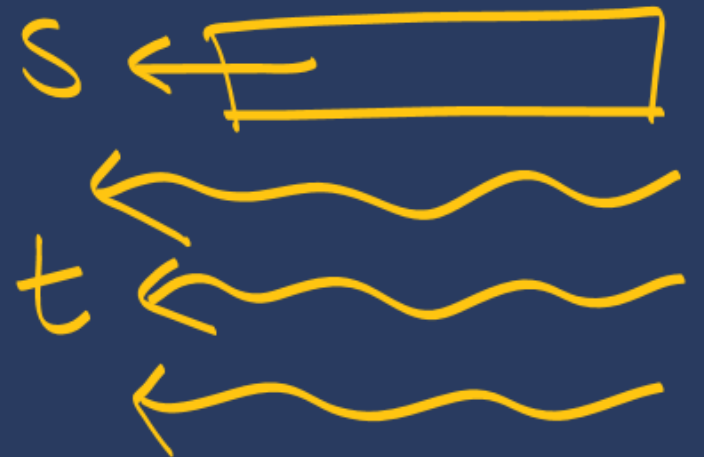


Speed of boat in
still water = s

Speed of stream = t

Net speed of boat $\Rightarrow (s - t)$
in upstream

DOWNSTREAM



Speed of boat in
still water = s

Speed of stream $\Rightarrow t$

Net speed in $\Rightarrow (s + t)$
downstream

#Upstream-Downstream:

#LP: A motorboat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream. (NCERT, 2020, 2018, AI 2014)

$$S = 18 \text{ km/hr.}$$

let t is speed of stream.

$$t_{up} = 1 + t_d$$

$$\Rightarrow \frac{d_u}{S_u} = 1 + \frac{d_d}{S_d}$$

$$\Rightarrow \frac{24}{S-t} = 1 + \frac{24}{S+t}$$

$$\Rightarrow \boxed{\frac{24}{18-t} = 1 + \frac{24}{18+t}}$$

#LP: Speed of a boat in still water is 15 km/h. It goes 30 km upstream and returns back at the same point in 4 hours 30 minutes. Find the speed of the stream. (Delhi 2017)

$$S = 15 \text{ km/hr}$$

$t \rightarrow$ let speed of stream.

Ans

$$\text{Total time} = 4.5 \text{ hr}$$

$$t_u + t_d = 4.5$$

$$\Rightarrow \frac{d_u}{S_u} + \frac{d_d}{S_d} = 4.5$$

$$\Rightarrow \frac{30}{S-t} + \frac{30}{S+t} = 4.5$$

$$\Rightarrow \frac{30}{15-t} + \frac{30}{15+t} = 4.5$$

$$d = s \times t$$

$$\frac{d}{s} = t$$

#LP: A 2-digit number is such that the product of its digits is 24. If 18 is subtracted from the number, the digits interchange their places. Find the number. (Term II, 2021-22)

Let unit place digit = y
 tens " " = x

$$N = 10x + y$$

Interchange
 digits

$$U = x$$

$$T = y$$

$$N' = 10y + x$$

$$A1Q \quad \boxed{x \cdot y = 24} \text{--- (I)}$$

$$A1Q \quad N - 18 = N'$$

$$(10x + y) - 18 = 10y + x$$

$$\Rightarrow 10x - x + y - 10y = 18$$

$$\Rightarrow 9x - 9y = 18$$

$$\div 9 (x - y) = 2$$

$$\boxed{x - y = 2} \text{--- (II)}$$

$$\rightarrow y = \frac{24}{x}$$

put in (I)

$$x - \frac{24}{x} = 2$$

$$\Rightarrow \frac{x^2 - 24}{x} = 2$$

$$\rightarrow x^2 - 24 = 2x$$

$$\Rightarrow \boxed{x^2 - 2x - 24 = 0}$$

$$x^2 - 2x - 24 = 0$$

$$x^2 - 6x + 4x - 24 = 0$$

$$x(x-6) + 4(x-6) = 0$$

$$(x-6)(x+4) = 0$$

$$x-6=0$$

$$x=6$$

$$6-y=2$$

$$y=6-2=4$$

$$x+4=0$$

$$x=-4$$

$$-4-y=2$$

$$-4-2=y$$

$$y=-6$$

$$p+q = -2$$

$$pq = -24$$

$$\begin{array}{r|l} 2 & 24 \\ \hline 2 & 12 \\ \hline 1 & 24 \end{array}$$

$$\begin{array}{r|l} 1 & 24 \\ \hline 3 & 8 \end{array}$$

Final Ans $\rightarrow N = 10x + y$
 $= 10(6) + 4$

$$N = 64$$

$$x = -4$$

$$y = -6$$

$$N = 10x + y$$

$$= 10(-4) + (-6)$$

$$= -40 - 6$$

$$N = -46$$

#LP: The sum of two numbers is 15, if the sum of their reciprocals is $\frac{3}{10}$, find the numbers.

Let one no. is x
 $\therefore$ other no. is $(15-x)$

Ans

$$\frac{1}{x} + \frac{1}{15-x} = \frac{3}{10}$$

$$\Rightarrow \frac{15-x+x}{x(15-x)} = \frac{3}{10}$$

$$\Rightarrow 15 \times 10 = 3x(15-x)$$

$$\Rightarrow 150 = 45x - 3x^2$$

$$\Rightarrow \boxed{3x^2 - 45x + 150 = 0}$$



two consecutive numbers $\Rightarrow$ let $(x), (x+1)$

two consec. odd numbers $\Rightarrow$

1 3

3 5

5 7

$\Rightarrow$ let $(x), (x+2)$

1	2
2	3
3	4
(x)	$(x+1)$

two consec. even numbers $\Rightarrow$

2 4

4 6

let $(x), (x+2)$

two conseq. multiples of 7 :- 7, 14

14, 21

let $(n), (n+7)$

#LP: There is a circular park of diameter 65 m as shown in the following figure, where AB is a diameter. An entry gate is to be constructed at a point P on the boundary of the park such that distance of P from A is 35 m more than the distance of P from B. Find distance of point P from A and B respectively.

Pytha

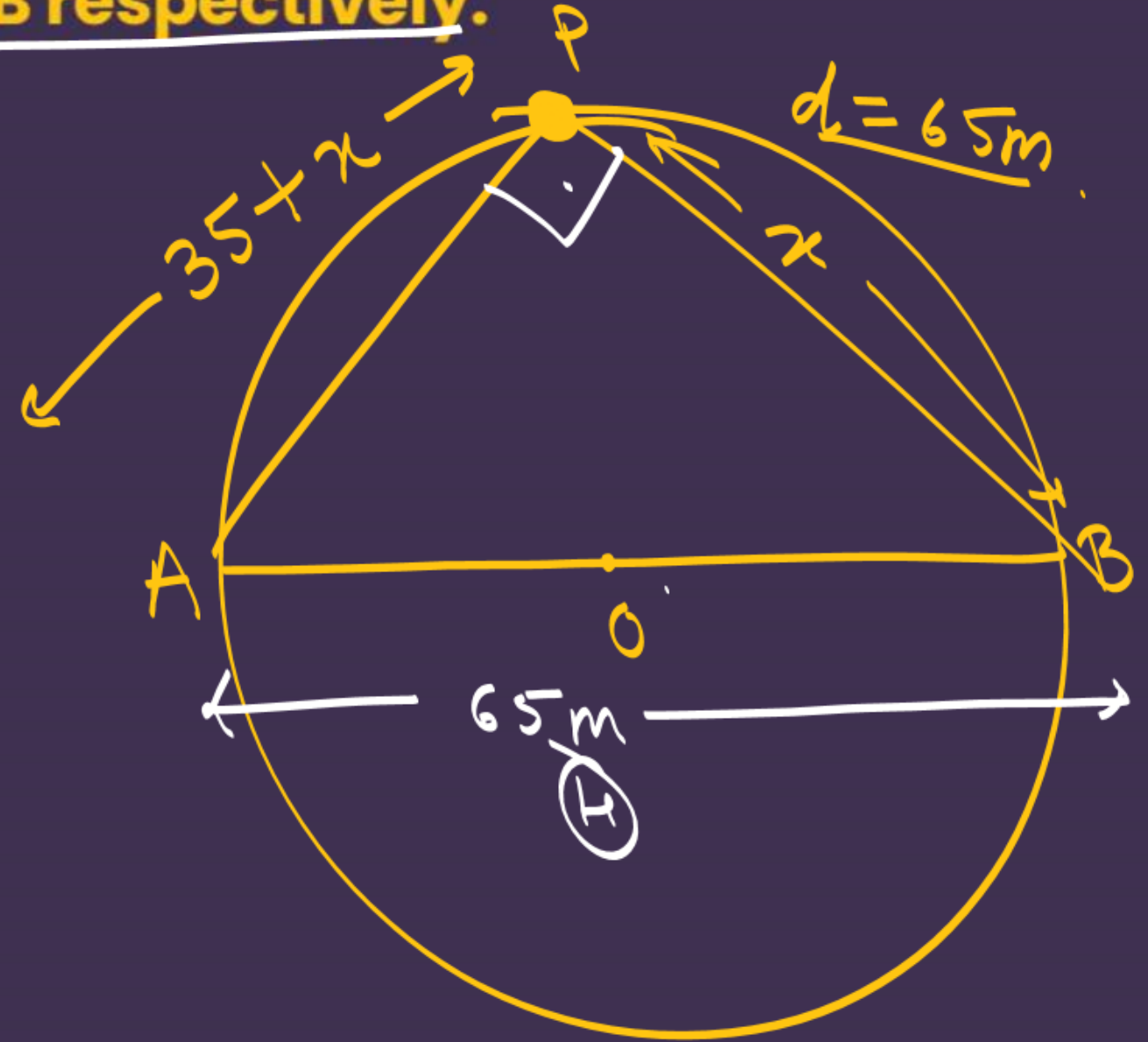
angle in semi-circle = 90

$$H^2 = P^2 + B^2$$

$$(65)^2 = (35+x)^2 + (x)^2$$

↓

ques.



#LP: Assertion: If $5+\sqrt{7}$ is a root of a quadratic equation with rational coefficients, then its other root is $5-\sqrt{7}$. ①

Reason: Surd roots of a quadratic equation with rational coefficients occur in conjugate pairs. ②

(A) ✓

$3+\sqrt{2}$
 $3-\sqrt{2}$ → conjugate pair

12/14 Completed!
Thank you Goodieess !!

